

“What are my chances of
success?”

This is me



Eating
fried
frog
legs





Start where you can

**Learning what the next step to take,
is just as important as taking it**



Ask for help

**It might surprise you just how many people are
willing to help if you ask nicely**



Come prepared

**Plan ahead which questions you want
to ask, and a mental list of topics you
would like to be asked about**



Remember success is a slide

**It takes a lot of hard work to achieve your goal, keep the
prize in mind when it gets tough to continue**

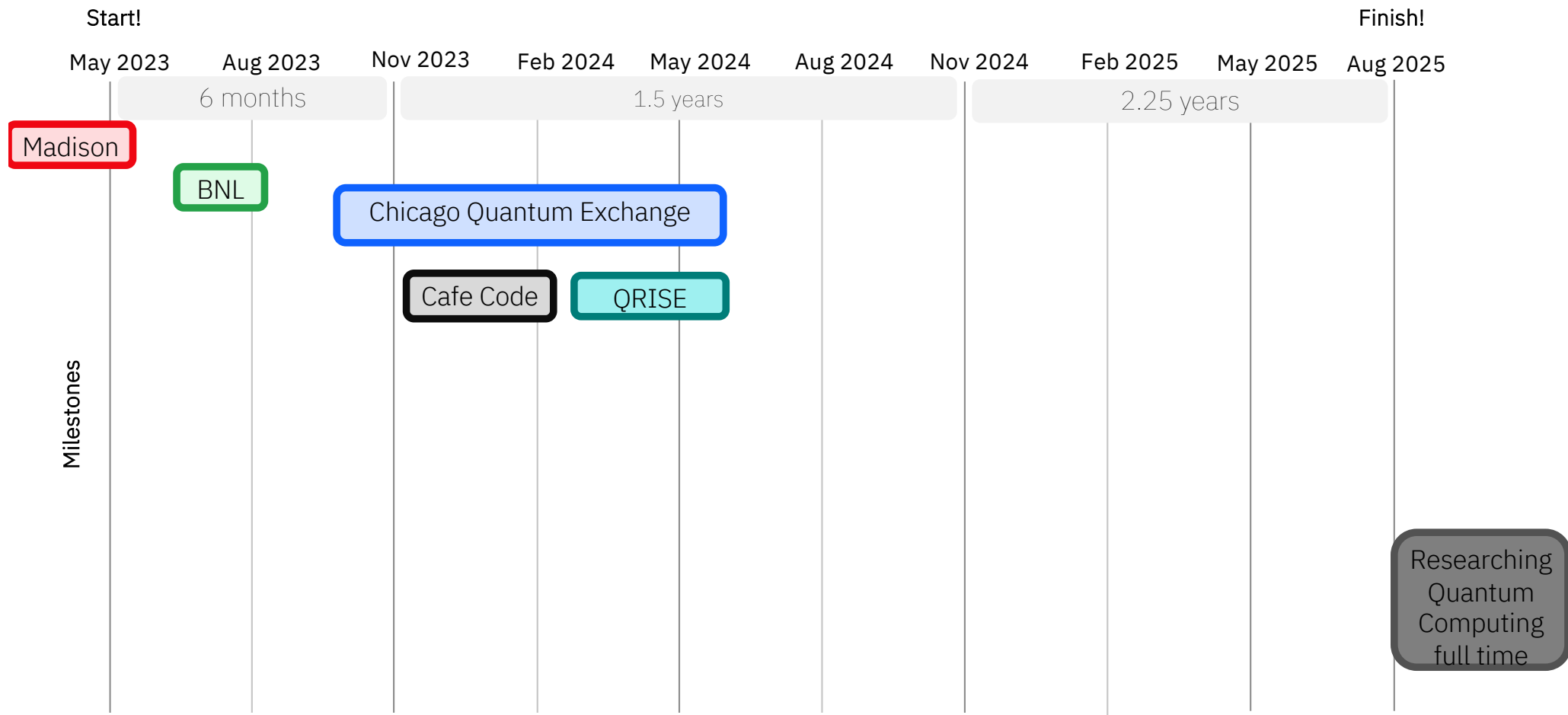
Start where you can

Choose projects
that progress the
goal

Search for
accessible
opportunities

Look to be
challenged, not
overwhelmed

Example:
QRISE Hackathon



Ask for help

Be proactive; find people who will help you

Own up if you don't understand; people who act smart get less help

Example:
GRFP application

Becky Durst

Mohammad Ramadan

Jeffrey Larson

Ji Liu

Siby Jose

Evan Toler

Wandi Ding

Sam Smiley

Emily Easton

Baha Balentekin

Matthew Tremba

Corrina Callahan

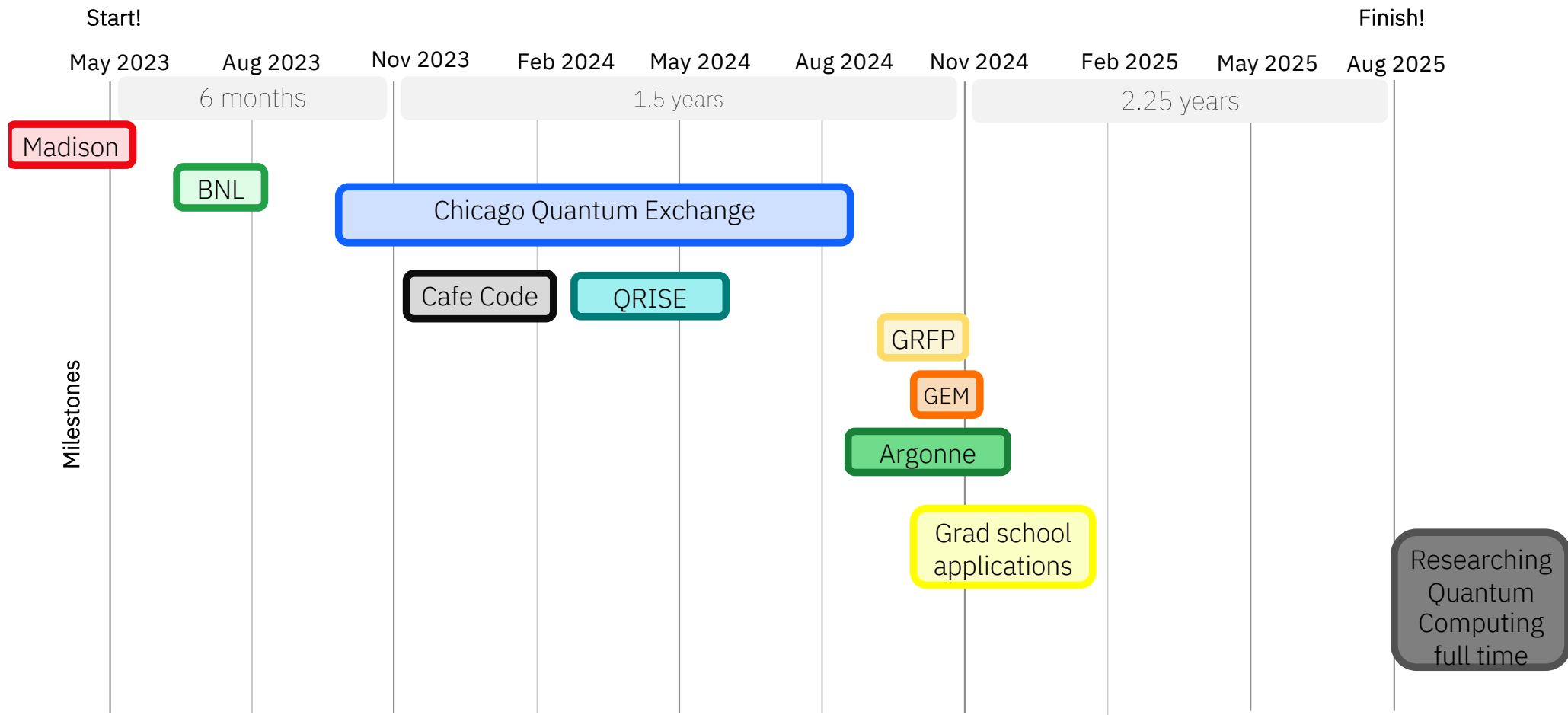
Bruce Lindvall

Josh Myers-dean

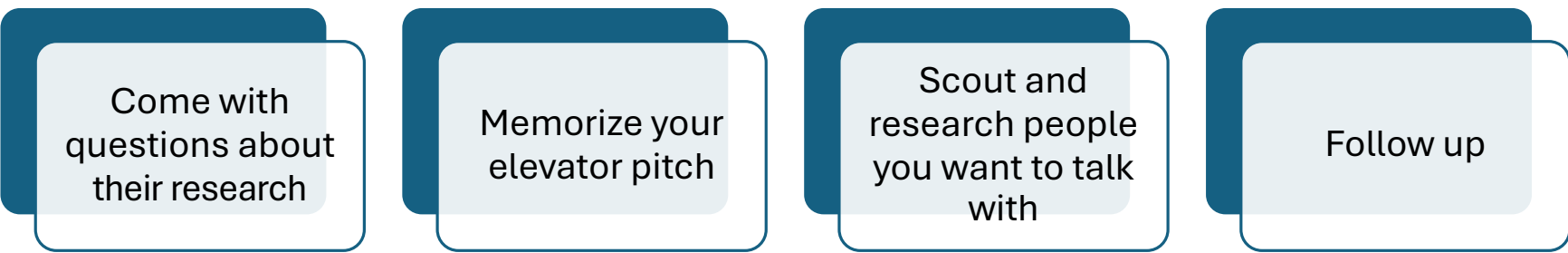
Amy Trethaway

Sridhar Tayur

Abhishek Aggarwal



Come prepared



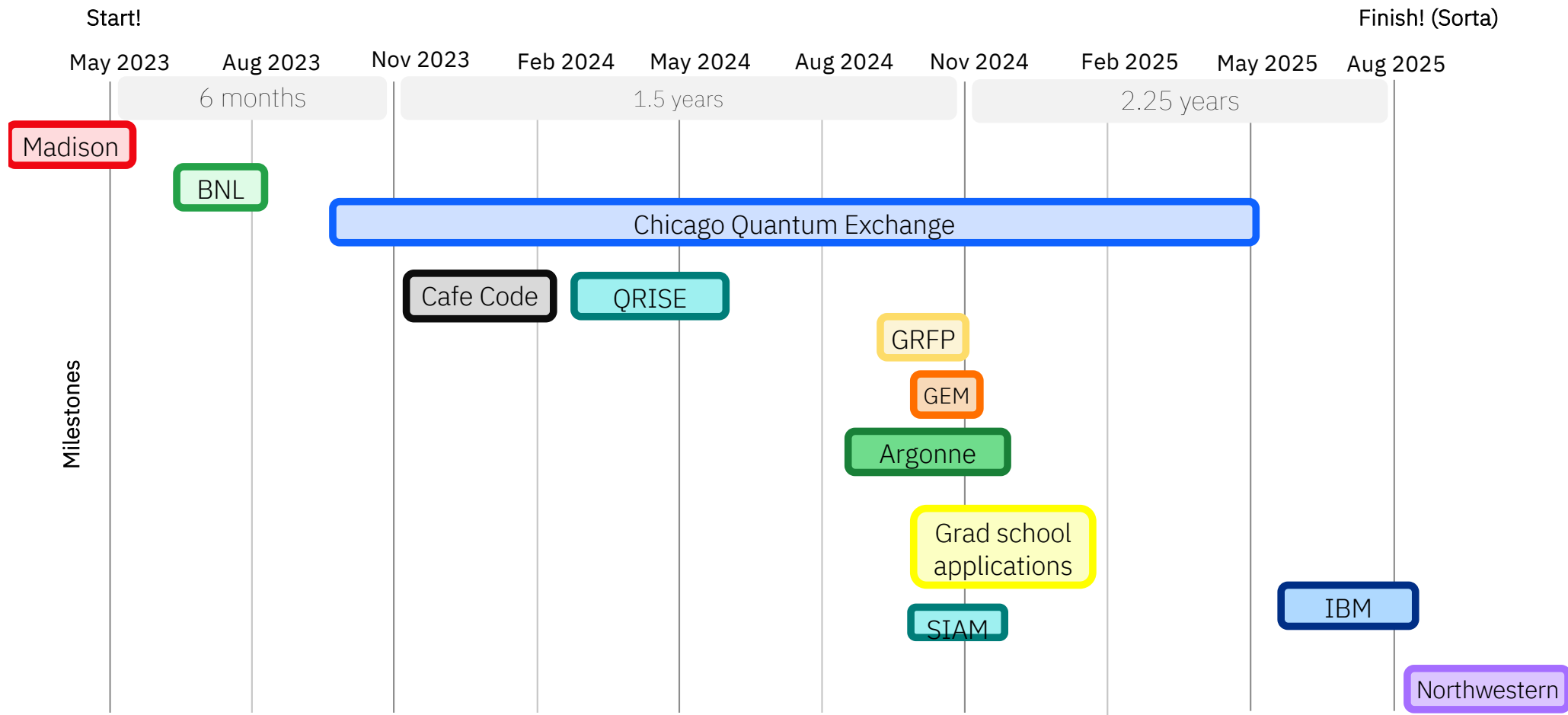
Come with
questions about
their research

Memorize your
elevator pitch

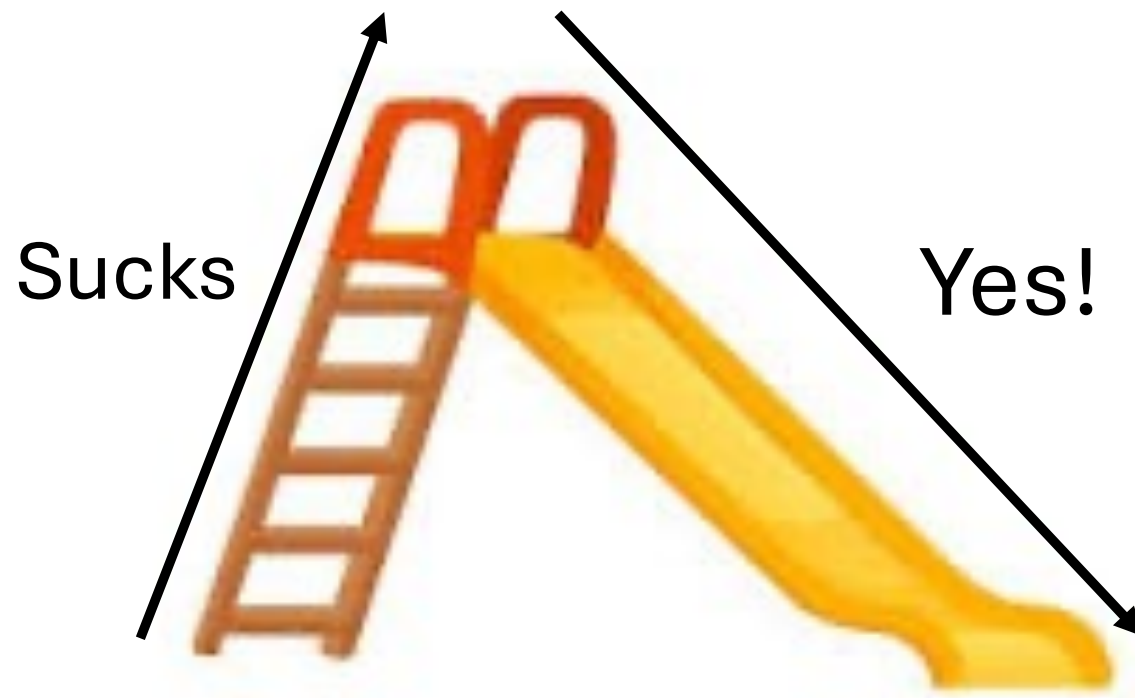
Scout and
research people
you want to talk
with

Follow up

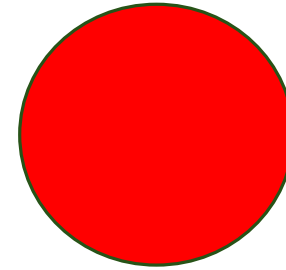
Example:
SIAM conference



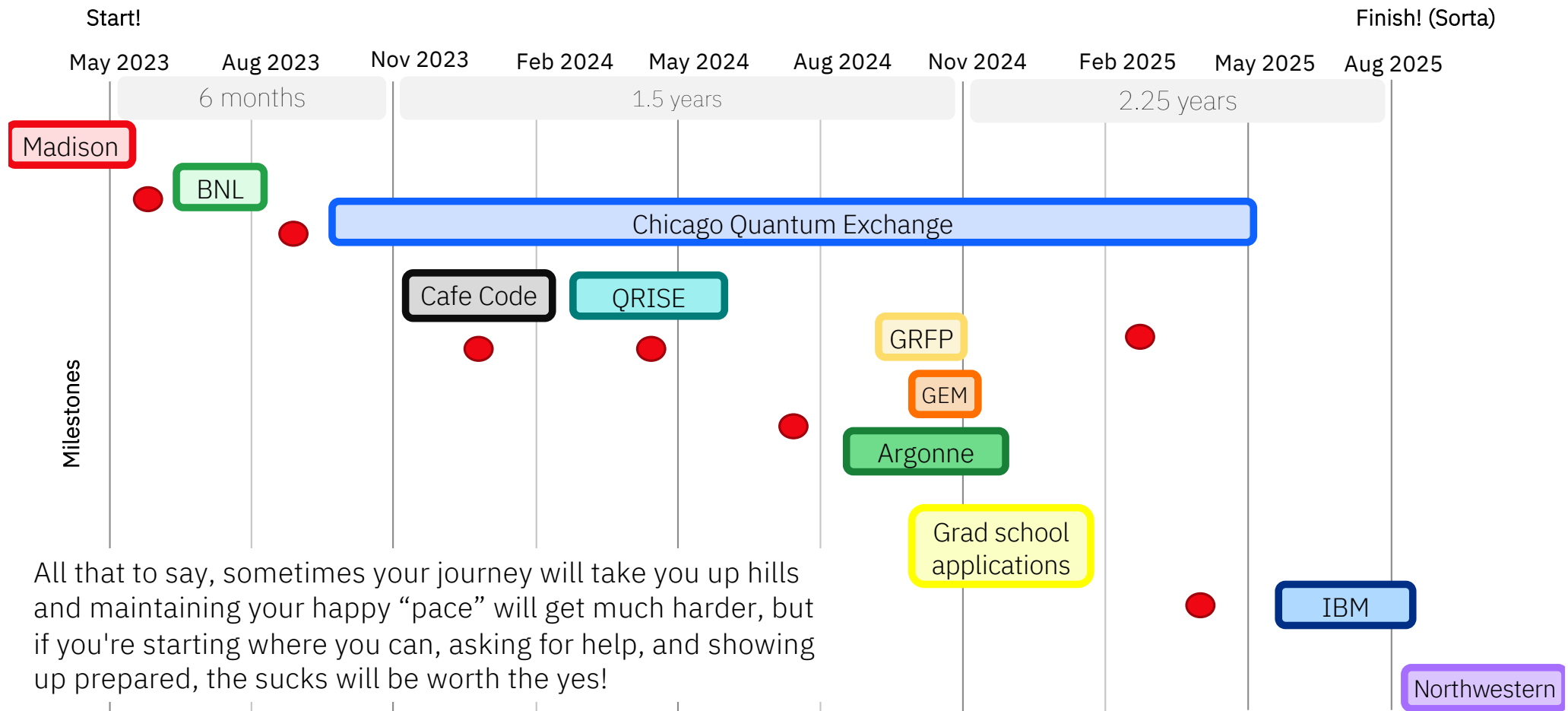
Remember success is a slide



The many moments of doubt



- 1. Got rejected from all grad programs I applied to
- 2. Believed I would never again be allowed to work in a STEM research role
- 3. Wanted to do research but didn't think I was smart enough
- 4. Seriously wondered if any of the work I was getting me any closer to my goal
- 5. Quit my full time job for an underpaid internship, hoping it would help me get into grad school
- 6. Application to GRFP rejected
- 7. Lost the GEM fellowship because of Northwestern funding cuts



Email me at:

Kaytlin.harrison@gmail.com

Thank You

Happy to connect on LinkedIn



Some example topics to ask about

- What is research in a National Lab is like
- How to get into university research
- Where to find **Scholarships and Grant funds**
- Research in Quantum Optimization (specifically QAOA)
- Transferring from a Community College
- Which Quantum Stocks to buy (just kidding!)
- Applying for the GRFP
- Applying STEM skills in a STEM **adjacent role**

Research at Argonne

Integration of Dynamic ADAPT-QAOA and QuCLEAR to reduce quantum gates in QAOA Circuits

Kaytlin Harrison, Ji Liu, Jeffrey Larson
Mathematics and Computer Science Division, Argonne National Laboratory

BACKGROUND

- **The Quantum Approximation Optimization Algorithm (QAOA)** solves QUBO problems like MaxCut by alternating between the Cost Hamiltonian H_C , which encodes the problem, and the Mixer Hamiltonian H_M which perturbs the system and lets us explore the solution space. These Hamiltonians use parameterized quantum gates which are optimized classically to minimize H_C
- A **key limitation** of QAOA is the high number of quantum gates required to encode the Hamiltonians[1]
- QuCLEAR and Dynamic ADAPT-QAOA (AQ) are unique optimization techniques that reduce the number of quantum gates needed for Standard QAOA

Dynamic ADAPT-QAOA

Dynamic AQ selects the mixer that will maximize the gradient, and determines is the H_C for a given layer is needed, optimizing the circuit by reducing the number of quantum gates required for convergence[2]

The energy variation due to the added parameters

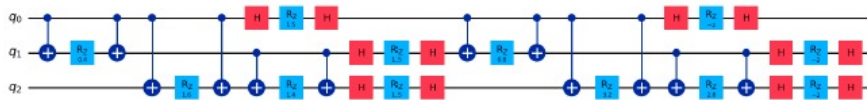
$$\delta E_n(\beta_n, \gamma_n; A) = \langle \Psi_n(\beta_n, \gamma_n; A) | H | \Psi_n(\beta_n, \gamma_n; A) \rangle$$

Enables the definition of the energy gradient

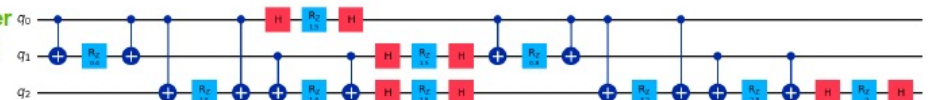
$$\mathcal{G}_p(\gamma_n; A) \equiv \left. \frac{\partial}{\partial \beta_n} \delta E_n(\beta_n, \gamma_n; A) \right|_{\beta_n=0}$$

Evaluating the gradient identifies the optimal mixer

$$M_n = \underset{A \in n}{\operatorname{argmax}} [|\mathcal{G}_n(\gamma_n; A)|]$$

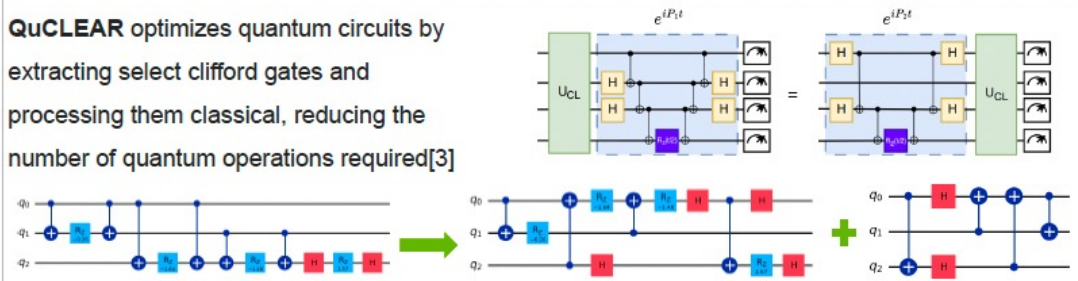


Before vs After
Dynamic AQ



QuCLEAR: QUANTUM CLIFFORD EXTRACTION AND ABSORPTION

QuCLEAR optimizes quantum circuits by extracting select clifford gates and processing them classical, reducing the number of quantum operations required[3]



MOTIVATION

- Will combining optimizations reduces quantum gates and error?
- Can Dynamic AQ and QuCLEAR work in tandem?
- Does the order of optimization matter?
- Is this a scalable solution?

METHODS

- Proof that order of optimization is inconsequential
- Empirical analysis of clifford gate counts for QAOA, Dynamic AQ, QuCLEAR, and their combination
- A look at scalability of both approaches combined

ORDER OF OPTIMIZATION

Initial setup: Layers of alternating Hc and Hm

Layer 0 = $|0\rangle$

Layer 0 + 1 = $M_1 C_1 |0\rangle$

$\sum_0^{n-1} \text{Layers} = M_{n-1} C_{n-1} \dots M_1 C_1 |0\rangle$

First: Use Dynamic AQ to determine the mixer M_n for layer n

$$M_n = \operatorname{argmax} \left[\left| \frac{\partial}{\partial \beta} \langle \psi_{n-1} | e^{-i\beta_n A} e^{-i\gamma_n H} | H | e^{-i\beta_n A} e^{-i\gamma_n H} | \psi_{n-1} \rangle \right| \right]$$

$\sum_0^n \text{Layers} = M_n C_n M_{n-1} C_{n-1} \dots M_1 C_1 |0\rangle$

Second: Use QuCLEAR to determine the optimized circuit and clifford circuit

$$\sum_0^n \text{Layers} = M_n C_n U_{CL} U_{\text{opt}} |0\rangle = U_{CL} M'_n C'_n U_{\text{opt}} |0\rangle$$

$$\text{where } M'_n = U_{CL}^\dagger M_n U_{CL} \quad C'_n = U_{CL}^\dagger C_n U_{CL}$$

Now: Reverse order and use QuCLEAR first to optimize layers 0 to $n-1$

$$\sum_0^{n-1} \text{Layers} = U_{CL} U_{\text{opt}} |0\rangle$$

Next: Use the QuCLEAR optimized layers and apply Dynamic AQ to get M_n

$$M_n = \operatorname{argmax} \left[\left| \frac{\partial}{\partial \beta} \langle 0 | U_{\text{opt}}^\dagger U_{CL}^\dagger e^{-i\beta_n A} e^{-i\gamma_n H} | H | e^{-i\beta_n A} e^{-i\gamma_n H} U_{CL} U_{\text{opt}} | 0 \rangle \right| \right]$$

$$M'_n = \operatorname{argmax} \left[\left| \frac{\partial}{\partial \beta} \langle 0 | U_{\text{opt}}^\dagger e^{-i\beta_n A'} e^{-i\gamma_n H'} U_{CL}^\dagger | H | U_{CL} e^{-i\beta_n A'} e^{-i\gamma_n H'} U_{\text{opt}} | 0 \rangle \right| \right]$$

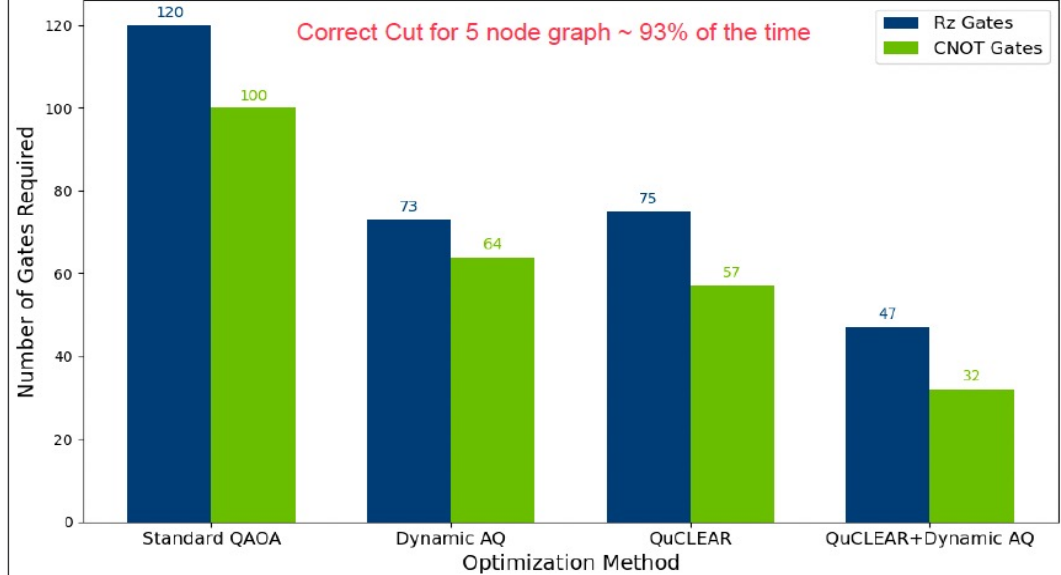
$$\sum_0^n \text{Layers} = U_{CL} M'_n C'_n U_{\text{opt}} |0\rangle$$

SCALABILITY

- In a 5-node example, QuCLEAR cut CNOT gates by 43% and RZ gates by 37.5%, while Dynamic AQ reduced them by 36% and 39%, respectively. Combined, they achieved reductions of 68% in CNOT gates and 60% in RZ gates
- But Dynamic AQ's gradient calculation introduces a computational bottleneck due to expensive matrix multiplications
- Inspired by Adapt-VQE, this is solved by calculating gradients in $O(n)$ time using FT-QC. Until such devices exist, we rely on costly classical methods for validation[4]

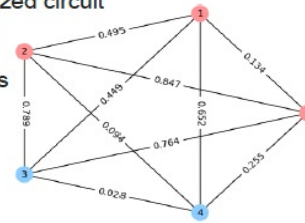
EMPIRICAL RESULTS

Comparison of Gate Counts Across Optimization Methods



SUMMARY

- Combining Dynamic AQ and QuCLEAR reduces quantum gates and sources of error
- The application order of techniques does not affect the final optimized circuit
- Dynamic AQ scales well with quantum gradient computations



FUTURE RESEARCH

Dynamic AQ's costliest step is gradient calculation. Future work could shrink the mixer pool or use machine learning to predict the best mixer, avoiding gradient evaluations

REFERENCES

[1] Mazumder, A., & Tayur, S. (2024). Five starter problems: Solving quadratic unconstrained binary optimization models on quantum computers. *arXiv:2401.08989*. [2] Yanakiev, N., et al. (2023). Dynamic-ADAPT-QAOA: An algorithm with shallow and noise-resilient circuits. *arXiv:2309.00047*. [3] Liu, J., et al. (2024). QuCLEAR: Clifford extraction and absorption for significant reduction in quantum circuit size. *arXiv:2408.13316*. [4] Zhu, L. An adaptive quantum approximate optimization algorithm for solving combinatorial problems on a quantum computer. *arXiv:2005.10258* (2022)